```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/IMDbMovies-Clean.csv")
```

```
df.head()
```

	Title	Summary	Director	Writer	Main Genres	Motion Picture Rating	Release Year	Runtime (Minutes)	Rating (Out of 10)	Number of Ratings (in thousands)	Budget (in millions)	Gross in US & Canada (in millions)	Gross worldwide (in millions)	Opening Weekend in US & Canada	Gross Opening Weekend (in millions)
0	Napoleon	An epic that details the checkered rise and fa...	Ridley Scott	David Scarpa	Action,Adventure,Biography	R	2023.0	158.0	6.7	38.0	NaN	37.514	84.968	11.26.2023	20.639
1	The Hunger Games: The Ballad of Songbirds & Sn...	Coriolanus Snow mentors and develops feelings ...	Francis Lawrence	Michael Lesslie,Michael Arndt,Suzanne Collins	Action,Adventure,Drama	PG-13	2023.0	157.0	7.2	37.0	100.0	105.043	191.729	11.19.2023	44.607
2	The Killer	After a fateful near-miss, an assassin battles...	David Fincher	Andrew Kevin Walker,Luc Jacamon,Alexis Nolent	Action,Adventure,Crime	R	2023.0	118.0	6.8	117.0	NaN	NaN	0.421	NaN	NaN
3	Leo	A 74-year-old lizard named Leo and his turtle ...	David Wachtenheim,Robert Smigel,Robert Marianetti	Paul Sado,Robert Smigel,Adam Sandler	Animation,Comedy,Family	PG	2023.0	102.0	7.0	10.0	NaN	NaN	NaN	NaN	NaN
4	Thanksgiving	After a Black Friday riot ends in tragedy, a m...	Eli Roth	Eli Roth,Jeff Rendell	Horror,Mystery,Thriller	R	2023.0	106.0	7.0	9.1	NaN	25.409	29.667	11.19.2023	10.306

Next steps: Generate code with df New interactive sheet

```
df.sample(2)
```

	Title	Summary	Director	Writer	Main Genres	Motion Picture Rating	Release Year	Runtime (Minutes)	Rating (Out of 10)	Number of Ratings (in thousands)	Budget (in millions)	Gross in US & Canada (in millions)	Gross worldwide (in millions)	Opening Weekend in US & Canada	Gross Opening Weekend (in millions)
4405	Always Be My Maybe	Everyone assumed that Sasha and Marcus would w...	Nahnatchka Khan	Randall Park,Ali Wong,Michael Golamco	Comedy,Romance	PG-13	2019.0	101.0	6.8	61.0	NaN	NaN	NaN	NaN	NaN
2375	Free Guy	When Guy, a bank teller, learns that he is a n...	Shawn Levy	Matt Lieberman,Zak Penn	Action,Adventure,Comedy	PG-13	2021.0	115.0	7.1	409.0	120.0	121.627	331.527	08.15.2021	28.365

```
df.describe()
```

	Release Year	Runtime (Minutes)	Rating (Out of 10)	Number of Ratings (in thousands)	Budget (in millions)	Gross in US & Canada (in millions)	Gross worldwide (in millions)	Gross Opening Weekend (in millions)
count	9076.000000	8918.000000	8813.00000	8813.000000	5879.000000	6064.000000	7128.000000	5695.000000
mean	2004.142133	107.022763	6.46613	125.215874	67.843117	46.809895	87.122463	12.449987
std	18.849816	25.369585	1.04979	208.132156	710.278383	72.906104	179.378745	22.363104
min	1915.000000	1.000000	1.30000	1.000000	0.000000	0.000000	0.000000	0.000000
25%	1994.000000	94.000000	5.80000	18.000000	6.000000	4.105000	2.923000	0.204500
50%	2010.000000	104.000000	6.50000	50.000000	20.000000	22.978500	23.838000	5.304000
75%	2019.000000	117.000000	7.20000	133.000000	46.150000	57.474500	88.971250	14.694000
max	2027.000000	607.000000	10.00000	2800.000000	30000.000000	936.662000	2923.706000	357.115000

```
df.shape
```

```
(9083, 15)
```

```
df.columns
```

```
Index(['Title', 'Summary', 'Director', 'Writer', 'Main Genres',
       'Motion Picture Rating', 'Release Year', 'Runtime (Minutes)',
       'Rating (Out of 10)', 'Number of Ratings (in thousands)',
       'Budget (in millions)', 'Gross in US & Canada (in millions)',
       'Gross worldwide (in millions)', 'Opening Weekend in US & Canada',
       'Gross Opening Weekend (in millions)'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9083 entries, 0 to 9082
Data columns (total 15 columns):
 #   Column                              Non-Null Count  Dtype
---  ------                              --------------  -----
 0   Title                               9083 non-null   object
 1   Summary                             9083 non-null   object
 2   Director                            9052 non-null   object
 3   Writer                              8759 non-null   object
 4   Main Genres                         9076 non-null   object
 5   Motion Picture Rating               8285 non-null   object
 6   Release Year                        9076 non-null   float64
 7   Runtime (Minutes)                   8918 non-null   float64
 8   Rating (Out of 10)                  8813 non-null   float64
 9   Number of Ratings (in thousands)    8813 non-null   float64
 10  Budget (in millions)                5879 non-null   float64
 11  Gross in US & Canada (in millions)  6064 non-null   float64
 12  Gross worldwide (in millions)       7128 non-null   float64
 13  Opening Weekend in US & Canada      5695 non-null   object
 14  Gross Opening Weekend (in millions) 5695 non-null   float64
dtypes: float64(8), object(7)
memory usage: 1.0+ MB
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	31
Writer	324
Main Genres	7
Motion Picture Rating	798
Release Year	7
Runtime (Minutes)	165
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	3204
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
#Data Cleaning
df["Main Genres"].mode()
```

	Main Genres
0	Drama

dtype: object

```
df["Main Genres"]= df["Main Genres"].fillna(df["Main Genres"]).mode()[0]
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	31
Writer	324
Main Genres	0
Motion Picture Rating	798
Release Year	7
Runtime (Minutes)	165
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	3204
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Director"] = df["Director"].fillna(df["Director"]).mode()[0]
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	324
Main Genres	0
Motion Picture Rating	798
Release Year	7
Runtime (Minutes)	165
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	3204
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Writer"]= df["Writer"].fillna(df["Writer"]).mode()[0]
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	798
Release Year	7
Runtime (Minutes)	165
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	3204
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Motion Picture Rating"].mode()
```

	Motion Picture Rating
0	R

dtype: object

```
df["Motion Picture Rating"]= df["Motion Picture Rating"].fillna(df["Motion Picture Rating"]).mode()[0]
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	7
Runtime (Minutes)	165
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	3204
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Release Year"]= df["Release Year"].fillna(df["Release Year"]).mean()
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	165
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	3204
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Runtime (Minutes)"].mean()
```

```
np.float64(107.02276295133439)
```

```
df["Runtime (Minutes)"]= df["Runtime (Minutes)"].fillna(df["Runtime (Minutes)"]).mean()
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	0
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	3204
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Budget (in millions)"]= df["Budget (in millions)"].fillna(df["Budget (in millions)"]).mean()
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	0
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	0
Gross in US & Canada (in millions)	3019
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Gross in US & Canada (in millions)"]= df["Gross in US & Canada (in millions)"].fillna(df["Gross in US & Canada (in millions)"]).mean()
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	0
Rating (Out of 10)	270
Number of Ratings (in thousands)	270
Budget (in millions)	0
Gross in US & Canada (in millions)	0
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Rating (Out of 10)"]= df["Rating (Out of 10)"].fillna(df["Rating (Out of 10)"]).mean()
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	0
Rating (Out of 10)	0
Number of Ratings (in thousands)	270
Budget (in millions)	0
Gross in US & Canada (in millions)	0
Gross worldwide (in millions)	1955
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Gross worldwide (in millions)"]= df["Gross worldwide (in millions)"].fillna(df["Gross worldwide (in millions)
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	0
Rating (Out of 10)	0
Number of Ratings (in thousands)	270
Budget (in millions)	0
Gross in US & Canada (in millions)	0
Gross worldwide (in millions)	0
Opening Weekend in US & Canada	3388
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df["Opening Weekend in US & Canada"]= df["Opening Weekend in US & Canada"].fillna(df["Opening Weekend in US & Canada"]).mode()[0]
```

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	0
Rating (Out of 10)	0
Number of Ratings (in thousands)	270
Budget (in millions)	0
Gross in US & Canada (in millions)	0
Gross worldwide (in millions)	0
Opening Weekend in US & Canada	0
Gross Opening Weekend (in millions)	3388

dtype: int64

```
df.drop(["Gross Opening Weekend (in millions)","Number of Ratings (in thousands)"], axis=1, inplace=True)
```

```
df.head()
```

	Title	Summary	Director	Writer	Main Genres	Motion Picture Rating	Release Year	Runtime (Minutes)	Rating (Out of 10)	Budget (in millions)	Gross in US & Canada (in millions)	Gross worldwide (in millions)	Opening Weekend in US & Canada
0	Napoleon	An epic that details the checkered rise and fa...	Steven Spielberg	John Hughes	Drama	R	2004.142133	107.022763	6.46613	67.843117	46.809895	87.122463	12.25.2016
1	The Hunger Games: The Ballad of Songbirds & Sn...	Coriolanus Snow mentors and develops feelings ...	Steven Spielberg	John Hughes	Drama	R	2004.142133	107.022763	6.46613	67.843117	46.809895	87.122463	12.25.2016
2	The Killer	After a fateful near-miss, an assassin battles...	Steven Spielberg	John Hughes	Drama	R	2004.142133	107.022763	6.46613	67.843117	46.809895	87.122463	12.25.2016
3	Leo	A 74-year-old lizard named Leo and his turtle ...	Steven Spielberg	John Hughes	Drama	R	2004.142133	107.022763	6.46613	67.843117	46.809895	87.122463	12.25.2016
4	Thanksgiving	After a Black Friday riot ends in tragedy, a m...	Steven Spielberg	John Hughes	Drama	R	2004.142133	107.022763	6.46613	67.843117	46.809895	87.122463	12.25.2016

Next steps: Generate code with df New interactive sheet

```
df.isnull().sum()
```

	0
Title	0
Summary	0
Director	0
Writer	0
Main Genres	0
Motion Picture Rating	0
Release Year	0
Runtime (Minutes)	0
Rating (Out of 10)	0
Budget (in millions)	0
Gross in US & Canada (in millions)	0
Gross worldwide (in millions)	0
Opening Weekend in US & Canada	0

dtype: int64

```
df["Profit"]= df["Gross worldwide (in millions)"]+df["Budget (in millions)"]
```

```
df.head()
```

	Title	Summary	Director	Writer	Main Genres	Motion Picture Rating	Release Year	Runtime (Minutes)	Rating (Out of 10)	Number of Ratings (in thousands)	Budget (in millions)	Gross in US & Canada (in millions)	Gross worldwide (in millions)	Opening Weekend in US & Canada	Gross Opening Weekend (in millions)	Profit
0	Napoleon	An epic that details the checkered rise and fa...	Ridley Scott	David Scarpa	Action,Adventure,Biography	R	2023.0	158.0	6.46613	38.0	NaN	37.514	84.968	11.26.2023	20.639	NaN
1	The Hunger Games: The Ballad of Songbirds & Sn...	Coriolanus Snow mentors and develops feelings ...	Francis Lawrence	Michael Lesslie,Michael Arndt,Suzanne Collins	Action,Adventure,Drama	PG-13	2023.0	157.0	6.46613	37.0	100.0	105.043	191.729	11.19.2023	44.607	291.729
2	The Killer	After a fateful near-miss, an assassin battles...	David Fincher	Andrew Kevin Walker,Luc Jacamon,Alexis Nolent	Action,Adventure,Crime	R	2023.0	118.0	6.46613	117.0	NaN	NaN	0.421	NaN	NaN	NaN
3	Leo	A 74-year-old lizard named Leo and his turtle ...	David Wachtenheim,Robert Smigel,Robert Marianetti	Paul Sado,Robert Smigel,Adam Sandler	Animation,Comedy,Family	PG	2023.0	102.0	6.46613	10.0	NaN	NaN	NaN	NaN	NaN	NaN
4	Thanksgiving	After a Black Friday riot ends in tragedy, a m...	Eli Roth	Eli Roth,Jeff Rendell	Horror,Mystery,Thriller	R	2023.0	106.0	6.46613	9.1	NaN	25.409	29.667	11.19.2023	10.306	NaN

```
df['rating_category'] = pd.cut(df['Rating (Out of 10)'],
bins = [0, 3, 6, 8, 10],
labels = ['Poor', 'Average', 'Good', 'Excellent'])
```

```
df["rating_category"].value_counts()
```

rating_category	count
Good	9083
Poor	0
Average	0
Excellent	0

dtype: int64

```
df['movie_category'] = pd.cut(df["Release Year"],
bins = [1929, 1979, 1989, 1999, 2020, 2027],
labels = ['Old is Gold', '80s Movies', '90s Movies', 'Mid Range', 'Latest New'])
```

```
df['movie_category'].value_counts()
```

movie_category	count
Mid Range	4267
Latest New	1905
90s Movies	1109
Old is Gold	985
80s Movies	796

dtype: int64

```
df.head()
```

	Title	Summary	Director	Writer	Main Genres	Motion Picture Rating	Release Year	Runtime (Minutes)	Rating (Out of 10)	Number of Ratings (in thousands)	Budget (in millions)	Gross in US & Canada (in millions)	Gross worldwide (in millions)	Opening Weekend in US & Canada	Gross Opening Weekend (in millions)	Profit	profit_status	rating_category	movie_category
0	Napoleon	An epic that details the checkered rise and fa...	Ridley Scott	David Scarpa	Action,Adventure,Biography	R	2023.0	158.0	6.7	38.0	NaN	37.514	84.968	11.26.2023	20.639	NaN	NaN	Good	Latest New
1	The Hunger Games: The Ballad of Songbirds & Sn...	Coriolanus Snow mentors and develops feelings ...	Francis Lawrence	Michael Lesslie,Michael Arndt,Suzanne Collins	Action,Adventure,Drama	PG-13	2023.0	157.0	7.2	37.0	100.0	105.043	191.729	11.19.2023	44.607	291.729	Hit	Good	Latest New
2	The Killer	After a fateful near-miss, an assassin battles...	David Fincher	Andrew Kevin Walker,Luc Jacamon,Alexis Nolent	Action,Adventure,Crime	R	2023.0	118.0	6.8	117.0	NaN	NaN	0.421	NaN	NaN	NaN	NaN	Good	Latest New
3	Leo	A 74-year-old lizard named Leo and his turtle ...	David Wachtenheim,Robert Smigel,Robert Marianetti	Paul Sado,Robert Smigel,Adam Sandler	Animation,Comedy,Family	PG	2023.0	102.0	7.0	10.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Good	Latest New
4	Thanksgiving	After a Black Friday riot ends in tragedy, a m...	Eli Roth	Eli Roth,Jeff Rendell	Horror,Mystery,Thriller	R	2023.0	106.0	7.0	9.1	NaN	25.409	29.667	11.19.2023	10.306	NaN	NaN	Good	Latest New

```
df['profit_status'] = pd.cut(df['Profit'],
bins = [-1,0, 500, 1000, 1500],
labels = ['Flop', 'Hit', 'Superhit', 'Blockbuster'])
```

```
df['profit_status'].value_counts()
```

profit_status	count
Hit	4940
Superhit	278
Blockbuster	74
Flop	0

dtype: int64

```
df["Runtime_Category"]= pd.qcut(df["Runtime (Minutes)"],3,
                                labels=["short","Medium","Long"])
```

```
df["Runtime_Category"].value_counts()
```

Runtime_Category	count
short	3006
Medium	2976
Long	2936

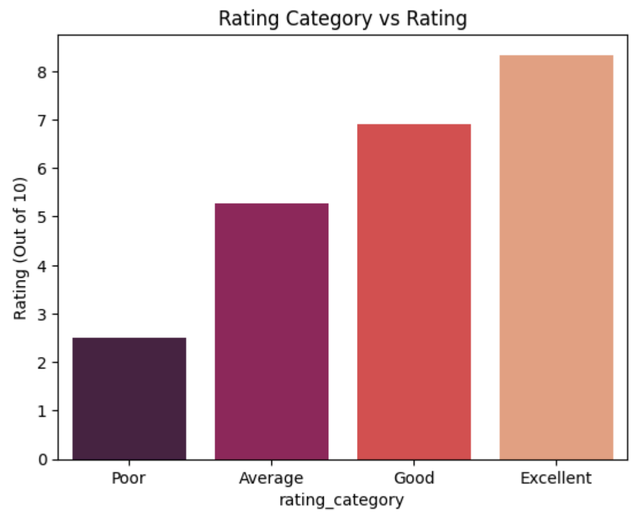
dtype: int64

```
df.head()
```

	Title	Summary	Director	Writer	Main Genres	Motion Picture Rating	Release Year	Runtime (Minutes)	Rating (Out of 10)	Number of Ratings (in thousands)	Budget (in millions)	Gross in US & Canada (in millions)	Gross worldwide (in millions)	Opening Weekend in US & Canada	Gross Opening Weekend (in millions)	Profit	profit_status	rating_category	movie_category	Runtime_Category
0	Napoleon	An epic that details the checkered rise and fa...	Ridley Scott	David Scarpa	Action,Adventure,Biography	R	2023.0	158.0	6.7	38.0	NaN	37.514	84.968	11.26.2023	20.639	NaN	NaN	Good	Latest New	Long
1	The Hunger Games: The Ballad of Songbirds & Sn...	Coriolanus Snow mentors and develops feelings ...	Francis Lawrence	Michael Lesslie,Michael Arndt,Suzanne Collins	Action,Adventure,Drama	PG-13	2023.0	157.0	7.2	37.0	100.0	105.043	191.729	11.19.2023	44.607	291.729	Hit	Good	Latest New	Long
2	The Killer	After a fateful near-miss, an assassin battles...	David Fincher	Andrew Kevin Walker,Luc Jacamon,Alexis Nolent	Action,Adventure,Crime	R	2023.0	118.0	6.8	117.0	NaN	NaN	0.421	NaN	NaN	NaN	NaN	Good	Latest New	Long
3	Leo	A 74-year-old lizard named Leo and his turtle ...	David Wachtenheim,Robert Smigel,Robert Marianetti	Paul Sado,Robert Smigel,Adam Sandler	Animation,Comedy,Family	PG	2023.0	102.0	7.0	10.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Good	Latest New	Medium
4	Thanksgiving	After a Black Friday riot ends in tragedy, a m...	Eli Roth	Eli Roth,Jeff Rendell	Horror,Mystery,Thriller	R	2023.0	106.0	7.0	9.1	NaN	25.409	29.667	11.19.2023	10.306	NaN	NaN	Good	Latest New	Medium

```
#EDA- Visualization
#Q1 - What is the Rating by movies

sns.barplot(data= df, x="rating_category", y="Rating (Out of 10)", palette="rocket" ,ci=False)
plt.title("Rating Category vs Rating")
```

```
/tmp/ipykernel_667/2578139408.py:3: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=('ci', False)` for the same effect.

  sns.barplot(data= df, x="rating_category", y="Rating (Out of 10)", palette="rocket" ,ci=False)
/tmp/ipykernel_667/2578139408.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data= df, x="rating_category", y="Rating (Out of 10)", palette="rocket" ,ci=False)
Text(0.5, 1.0, 'Rating Category vs Rating')
```



```
#Q2 What is distribute of movies profit
df["Profit"].value_counts()
```

Profit	count
3.114	3
17.500	3
47.485	3
2.000	3
10.019	2
...	...
0.940	1
9.524	1
34.702	1
41.288	1
272.290	1

5251 rows × 1 columns

dtype: int64

```
#Q3- what is movies Rating category by profit
```

```
df.groupby("rating_category")["Profit"].mean()
```

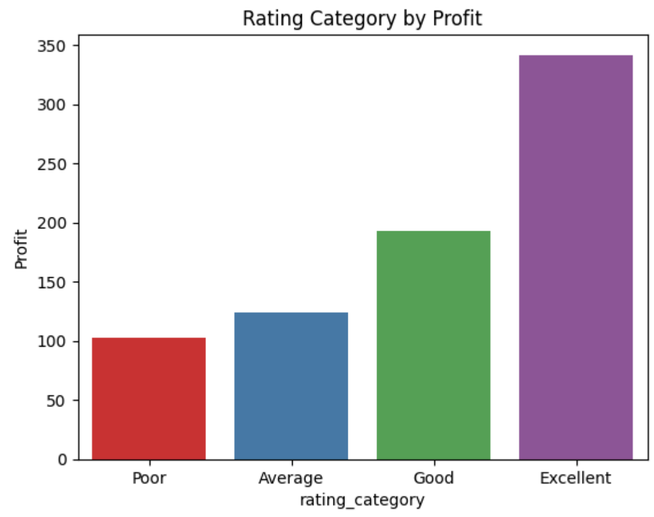
```
/tmp/ipykernel_541/3231873011.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this war
  df.groupby("rating_category")["Profit"].mean()
```

rating_category	Profit
Poor	102.796600
Average	123.938241
Good	192.780248
Excellent	341.783012

dtype: float64

```
sns.barplot(data= df, x="rating_category", y="Profit", palette="Set1",ci=False)
plt.title("Rating Category by Profit")
```

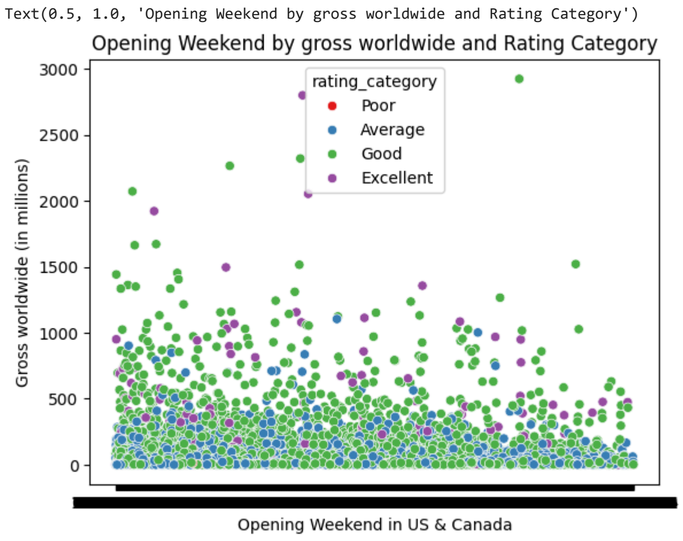
```
/tmp/ipykernel_541/1967249452.py:1: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=('ci', False)` for the same effect.

  sns.barplot(data= df, x="rating_category", y="Profit", palette="Set1",ci=False)
/tmp/ipykernel_541/1967249452.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data= df, x="rating_category", y="Profit", palette="Set1",ci=False)
Text(0.5, 1.0, 'Rating Category by Profit')
```




```
#Q4- what is Opening weekend US & Canada of gross in rating category ?
```

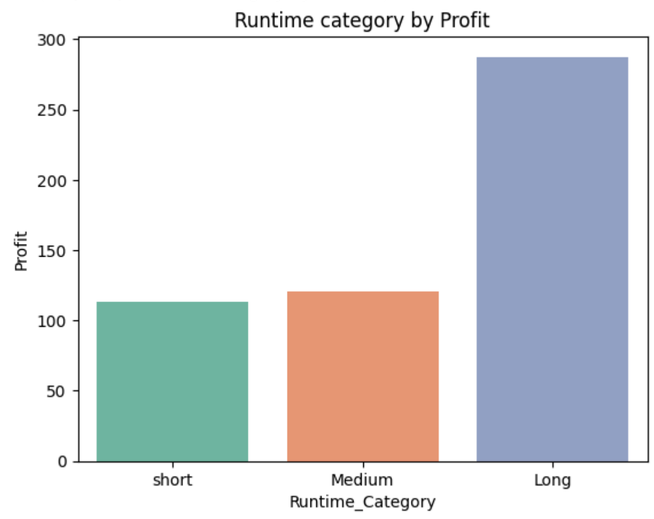
```
sns.scatterplot(data=df, x="Opening Weekend in US & Canada", y="Gross worldwide (in millions)", hue="rating_category", palette="Set1")
plt.title("Opening Weekend by gross worldwide and Rating Category" )
```

```
Text(0.5, 1.0, 'Opening Weekend by gross worldwide and Rating Category')
```



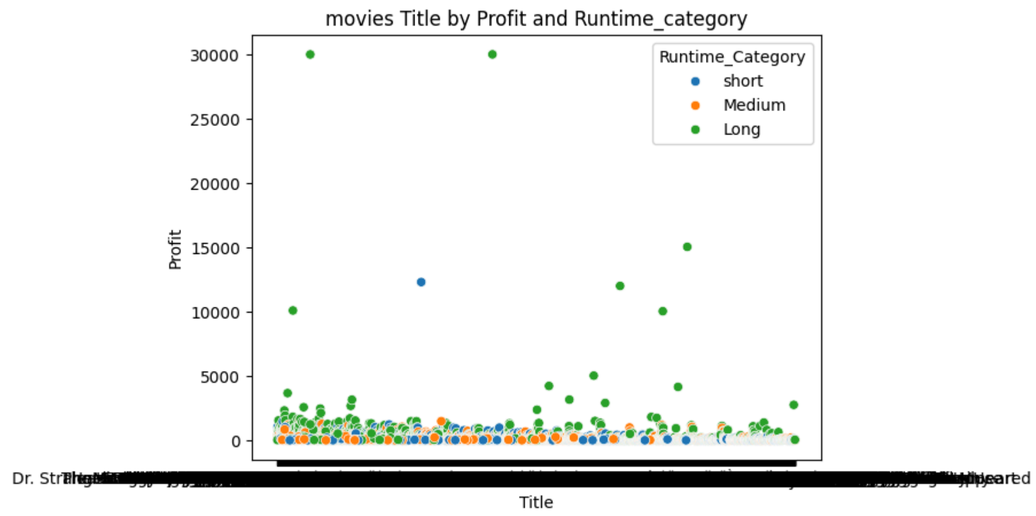

```
sns.barplot(data=df, x="Runtime_Category", y="Profit",palette="Set2",ci=False)
plt.title("Runtime category by Profit")
```

```
/tmp/ipykernel_541/2501148553.py:1: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=('ci', False)` for the same effect.

  sns.barplot(data=df, x="Runtime_Category", y="Profit",palette="Set2",ci=False)
/tmp/ipykernel_541/2501148553.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=df, x="Runtime_Category", y="Profit",palette="Set2",ci=False)
Text(0.5, 1.0, 'Runtime category by Profit')
```



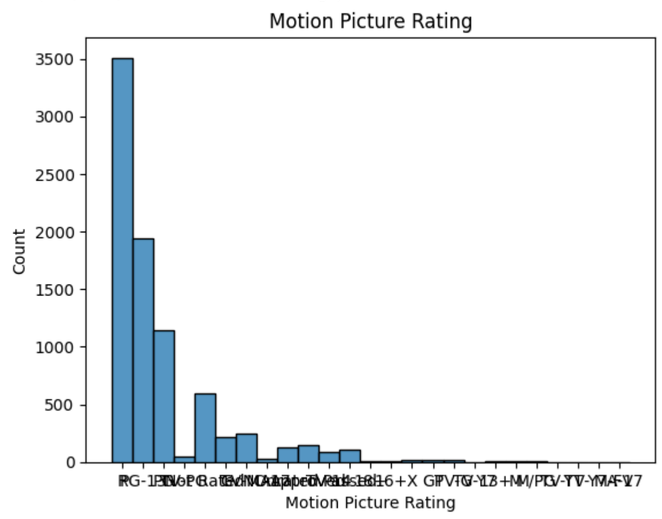
```
#5- which are the movies title and profit and runtime category
sns.scatterplot(data= df, x="Title", y="Profit", hue="Runtime_Category")
plt.title("movies Title by Profit and Runtime_category")
```

```
Text(0.5, 1.0, 'movies Title by Profit and Runtime_category')
```



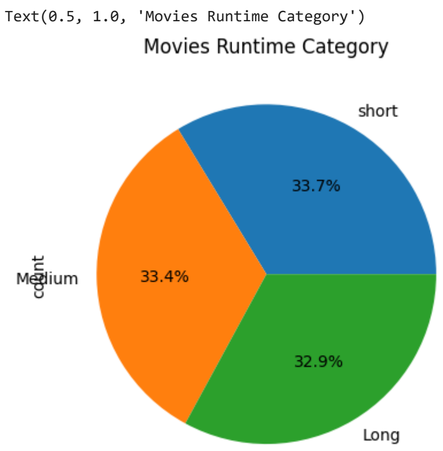

```
#Q6- what is the Motion picture Rating
sns.histplot(data=df, x="Motion Picture Rating",bins=30)
plt.title("Motion Picture Rating")
```

```
Text(0.5, 1.0, 'Motion Picture Rating')
```




```
#Q7- what is movies runtime category
df["Runtime_Category"].value_counts().plot.pie(autopct="%1.1f%%")
plt.title("Movies Runtime Category")
```

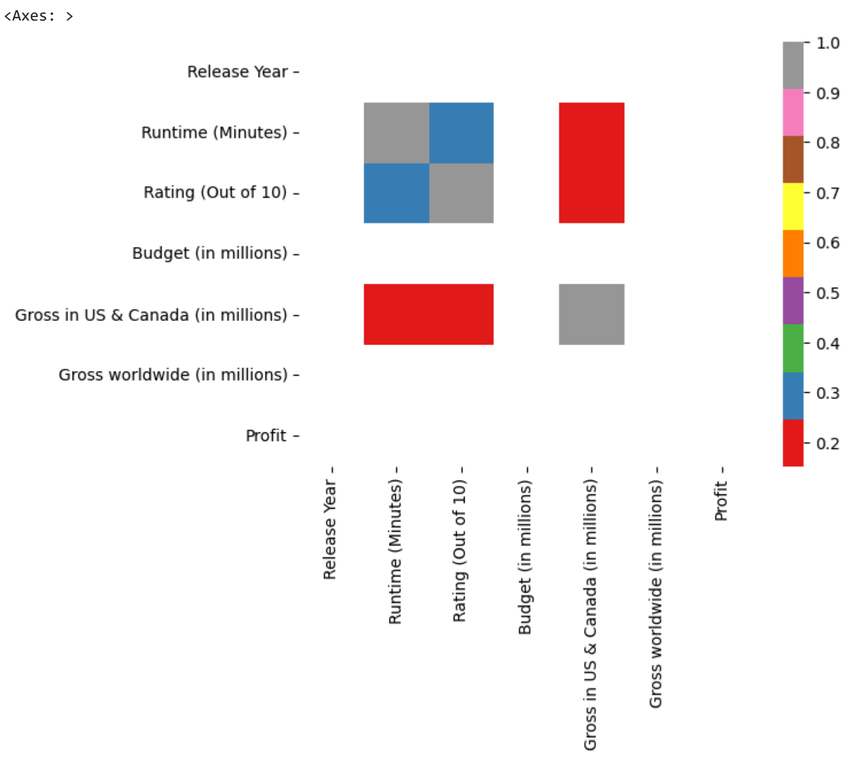
```
Text(0.5, 1.0, 'Movies Runtime Category')
```




```
correlation_matrix= df.corr(numeric_only= True)
```

```
sns.heatmap(data=correlation_matrix,cmap="Set1")
```

```
<Axes: >
```




```
#Q8- what is profit status by rating
df.groupby("profit_status")["Rating (Out of 10)"].mean()
```

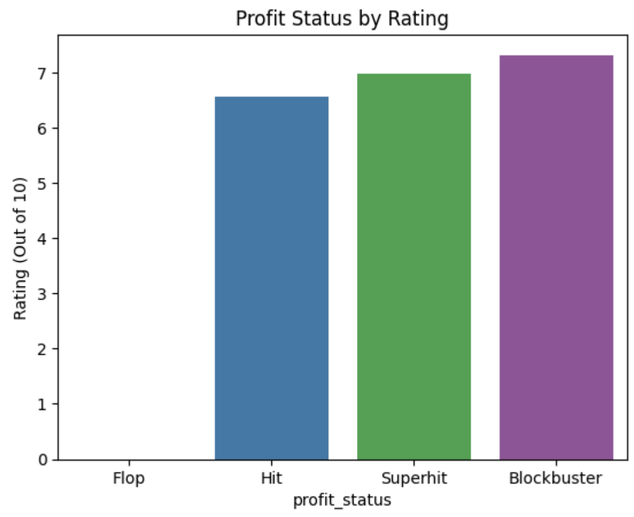
```
/tmp/ipykernel_4083/1275825661.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this wa
  df.groupby("profit_status")["Rating (Out of 10)"].mean()
```

profit_status	Rating (Out of 10)
Flop	NaN
Hit	6.550567
Superhit	6.975540
Blockbuster	7.312162

dtype: float64

```
sns.barplot(data=df, x="profit_status", y="Rating (Out of 10)", ci=False,palette="Set1")
plt.title("Profit Status by Rating")
```

```
/tmp/ipykernel_4083/3391194808.py:1: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=('ci', False)` for the same effect.

  sns.barplot(data=df, x="profit_status", y="Rating (Out of 10)", ci=False,palette="Set1")
/tmp/ipykernel_4083/3391194808.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=df, x="profit_status", y="Rating (Out of 10)", ci=False,palette="Set1")
Text(0.5, 1.0, 'Profit Status by Rating')
```




```
#Q9- what is the movie category by profit
df.groupby("movie_category")["Profit"].mean()
```

```
/tmp/ipykernel_4083/1405157050.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this wa
  df.groupby("movie_category")["Profit"].mean()
```

movie_category	Profit
Old is Gold	45.409508
80s Movies	119.463525
90s Movies	126.624359
Mid Range	207.439538
Latest New	360.828843

dtype: float64

```
sns.barplot(data=df, x="movie_category",y="Profit", ci=False, palette="Set1")
```